



North South University

Department of Electrical & Computer Engineering

Lab Report

Experiment No:	04
Experiment Title:	BCD to Excess-3 Converter
Course Code:	CSE231L
Section:	10
Course Name:	Digital Logic Design Lab
Lab Group #:	07
Written By:	Md Mafiul Haque (2322723642)
Date of Experiment:	18-10-2025
Date of Submission:	25-10-2025

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Experiment name: BCD to Excess-3 converter.

Objectives:

- i) Designing a complete minimal combinational logic system from specification to implementation.
- ii) Minimize combinational logic circuits using Karnaugh maps.
- iii) Learn various numerical representation systems.
- iv) Implement circuits using canonical minimal forms.

List of Equipments:

- i) Trainer board.
- ii) 1 x IC 4073 Triple 3-input AND gates.
- iii) 1 x IC 4075 Triple 3-input OR gates.
- iv) 1 x IC 7404 Hex inverters (Not gate).
- v) 1 x IC 7400 2-input NAND gates.
- vi) 2 x IC 7408 2-input AND gates.
- vii) 1 x IC 7432 Quadruple 2-input OR gates.

Theory:

BCD is a way of representing decimal numbers using binary digits. In BCD, each decimal digit is representing by a four bit binary code. Allowing each BCD digits to have one of 16 possible values (0000 to 1111). BCD is commonly used in digital systems where decimal arithmetics is required. Such as in calculators, counters and displays.

Excess-3 is a decimal code that represents decimal numbers using binary digits by adding 3 to the corresponding binary value of each decimal digit.

Excess-3 is commonly used in digital systems where decimal arithmetics is required such as in digital displays, calculators etc.

A K-map is a diagram made up of squares, with each square representing one minterm of function that is to be minimized. In fact, this represents a visual diagram of all possible ways a function may be expressed in standard form. By recognizing various patterns, it is possible to derive alternative algebraic expressions for the same function, from which the simplest can be selected.

W	X	Y	Z	A
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1

Table B1

Fig-B1

W \ YZ	00	01	11	10
00	m ₀ w'x'y'z'	m ₁ w'x'y'z	m ₃ w'x'y'z	m ₂ w'x'y'z'
01	m ₄ w'x'y'z	m ₅ w'x'y'z	m ₇ w'x'y'z	m ₆ w'x'y'z'
11	m ₁₂ wx'y'z'	m ₁₃ wx'y'z	m ₁₅ wx'y'z	m ₁₄ wx'y'z'
10	m ₈ wx'y'z'	m ₉ wx'y'z	m ₁₁ wx'y'z	m ₁₀ wx'y'z'

Fig-B2

W \ YZ	00	01	11	10
00	m ₀ 1	m ₁ 1	m ₃ 0	m ₂ 0
01	m ₄ 0	m ₅ 0	m ₇ 0	m ₆ 1
11	m ₁₂ X	m ₁₃ X	m ₁₅ X	m ₁₄ X
10	m ₈ 1	m ₉ 1	m ₁₁ X	m ₁₀ X

Figure B1 shows the minterm positions on the K-map of 4-input variables and 1 output variable.

Table B1 shows the truth table for the function

$A = WX + XYZ' + WZ$ where $WXYZ$ goes from binary 0 to 9.

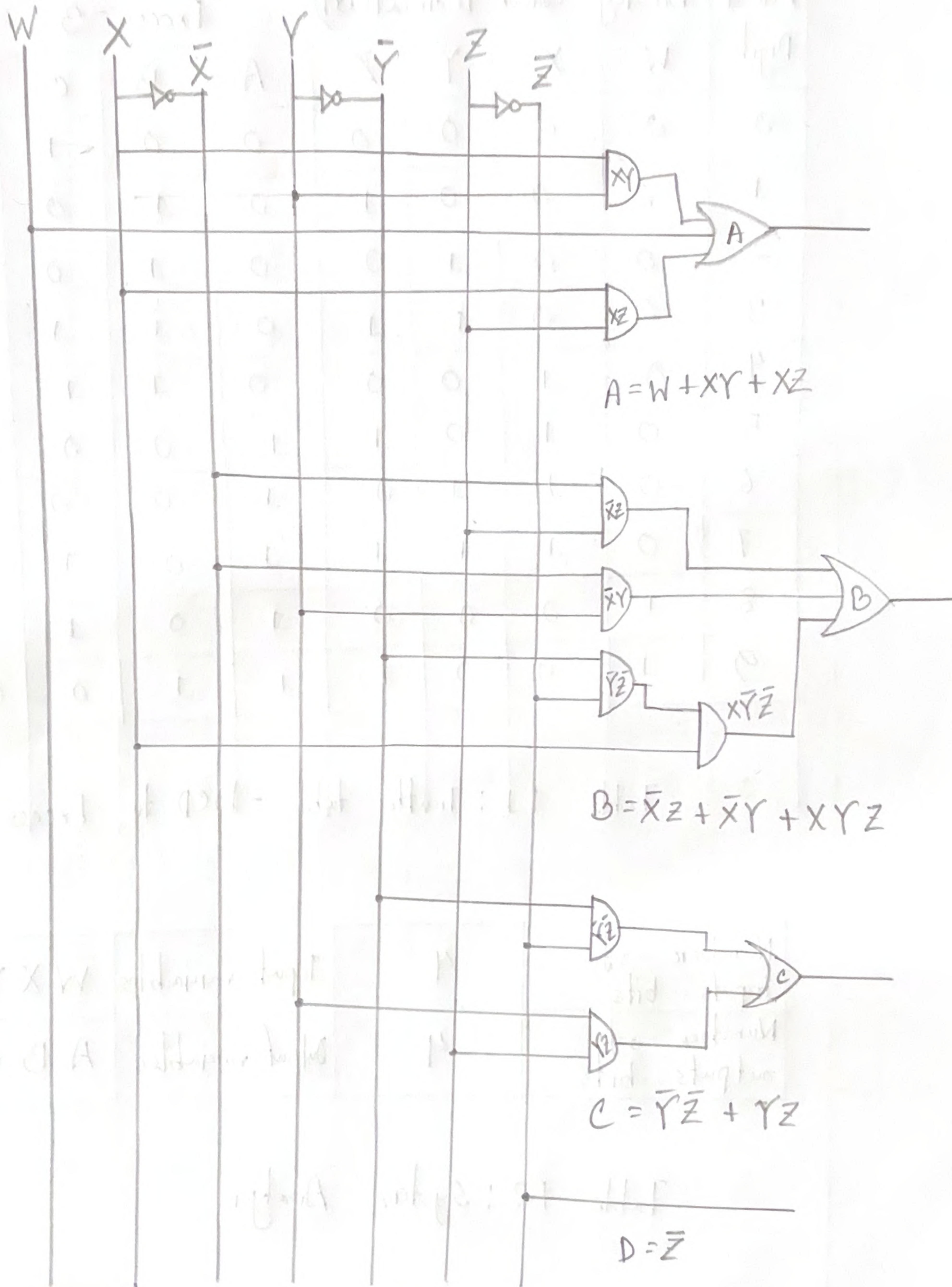
Figure B2 shows the K-map for the function that

we can use to minimize the function to

$$X'Y' + XYZ'$$

A	W	X	Y	Z
0	0	0	0	0
1	1	0	0	0
2	0	1	0	0
3	1	1	0	0
4	0	0	1	0
5	1	0	1	0
6	0	1	1	0
7	1	1	1	0
8	0	0	0	1
9	1	0	0	1

Circuit Diagram:



Experimental Data:

Decimal Digit	Binary Coded Decimal (BCD)				Excess - 3			
	W	X	Y	Z	A	B	C	D
0	0	0	0	0	0	0	1	1
1	0	0	0	1	0	1	0	0
2	0	0	1	0	0	1	0	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	0	1
7	0	1	1	1	1	0	1	0
8	1	0	0	0	1	0	1	1
9	1	0	0	1	1	1	0	0

Table F1 : Truth table - BCD to Excess-3

Number of inputs bits	4	Input variables	W X Y Z
Number of outputs bits	4	Output variables	A B C D

Table F2 : System Analysis

$wx \backslash yz$	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

$$A = W + XY + XZ$$

$wx \backslash yz$	00	01	11	10
00	0	1	1	1
01	1	0	0	0
11	X	X	X	X
10	0	1	X	X

$$B = \bar{X}Z + \bar{X}Y + X\bar{Y}\bar{Z}$$

$wx \backslash yz$	00	01	11	10
00	1	0	1	0
01	1	0	1	0
11	X	X	X	X
10	1	0	X	X

$$C = \bar{Y}\bar{Z} + YZ$$

$wx \backslash yz$	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	X	X	X	X
10	1	0	X	X

$$D = \bar{Z}$$

Figure f1: K-maps

Questions and Answers:

1) The IC diagram for output B of the BCD to Excess-3 converter has been drawn below:

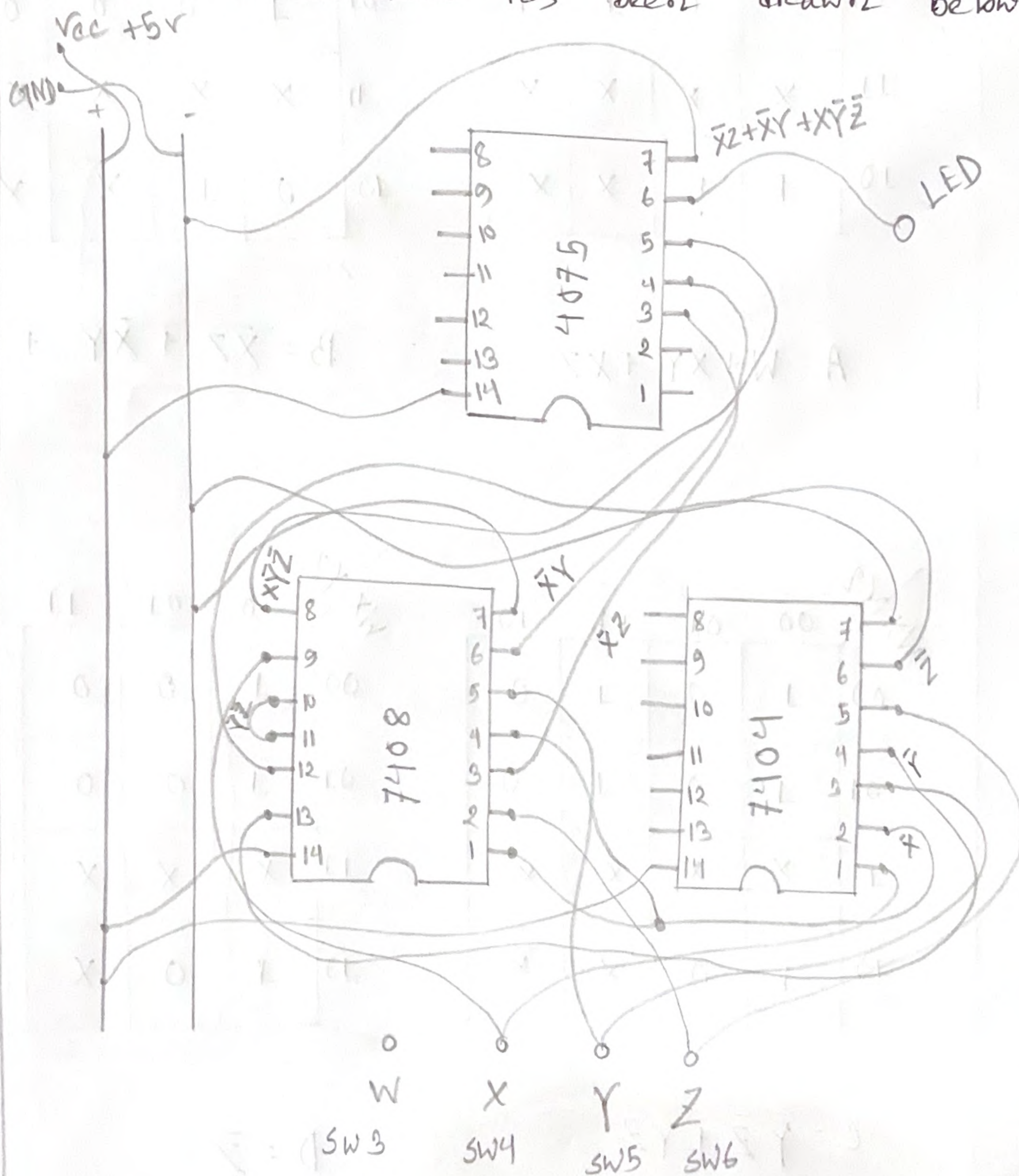


Figure: Diagram for output B of the BCD to Excess-3 converter.

2] The screenshots of Logisim circuit schematic and truth table for the entire BCD to Excess-3 converter has been provided.

Discussion:

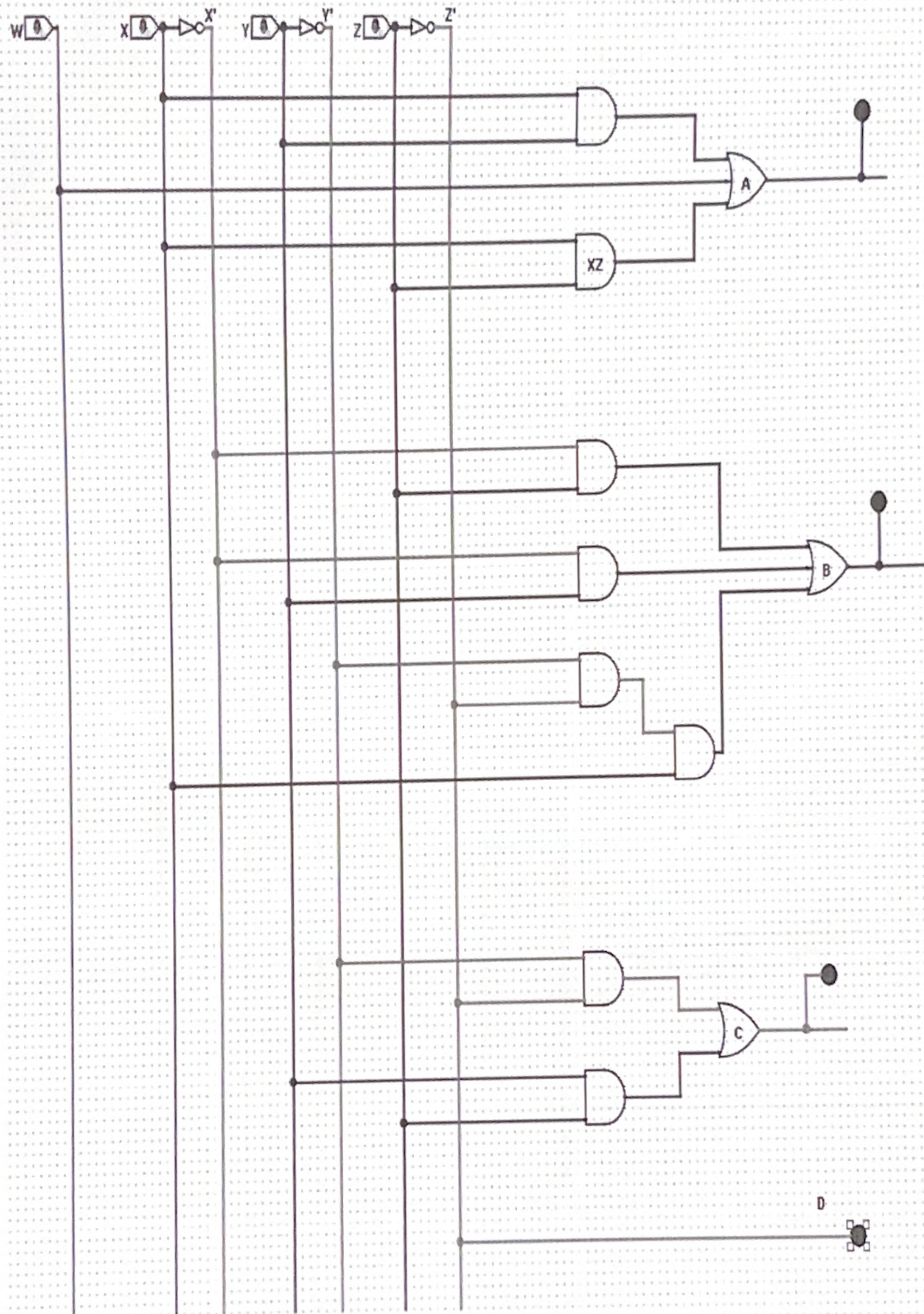
In this lab, we successfully built a circuit that converts from BCD to Excess-3. We learnt how to design a complete and minimal logic circuit for the converter.

To make our circuit as efficient as possible, we used K-map let us take the complex logic from our truth table and find the simplest possible boolean equation for each outputs (A, B, C, D)

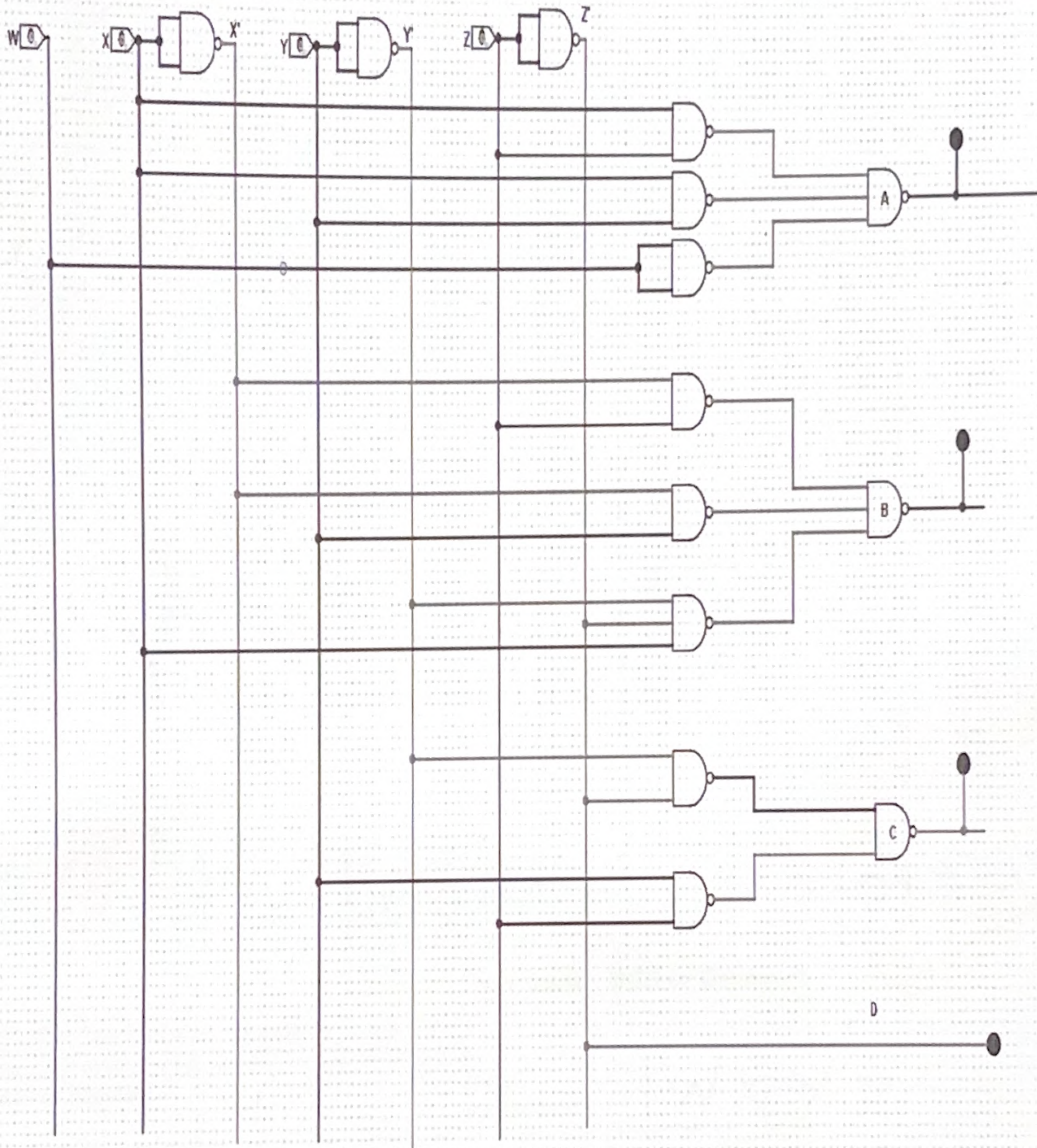
We built the circuit using the simple equation twice. First, we used a mixture of AND, NOT and OR gates and after that we used NAND gate to rebuild the circuit. This proved NAND gates are universal and reduced the number of IC needed for the circuit.

We also ran into a few problems. The 3-input OR gate (4075) was broken. So, we had to troubleshoot using 2-input OR gate and adapt to get the circuit working. This taught us that real hardware involves problem solving when parts fail. Overall, we learned a lot about K-maps, BCD code and designing and building efficient digital circuits.

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Bonus — 0.5

[Signature]

F. Experimental Data

Group: 07	Section: 10	Instructor's Signature: _____ Date: 18-10-2025
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Decimal Digit	Binary Coded Decimal (BCD)				Excess-3			
	W	X	Y	Z	A	B	C	D
0	0	0	0	0	0	0	1	1
1	0	0	0	1	0	1	0	0
2	0	0	1	0	0	1	0	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	0	1
7	0	1	1	1	1	0	1	0
8	1	0	0	0	1	0	1	1
9	1	0	0	1	1	1	0	0

Table F1: Truth table - BCD to Excess-3

Number of inputs bits:	4	Input variables:	W X Y Z
Number of outputs bits:	4	Output variables:	A B C D

Table F2: System analysis

WX \ YZ	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

WX \ YZ	00	01	11	10
00	0	1	1	1
01	1	0	0	0
11	X	X	X	X
10	0	1	X	X

$A = W + XZ + XY$

WX \ YZ	00	01	11	10
00	1	0	1	0
01	1	0	1	0
11	X	X	X	X
10	1	0	X	X

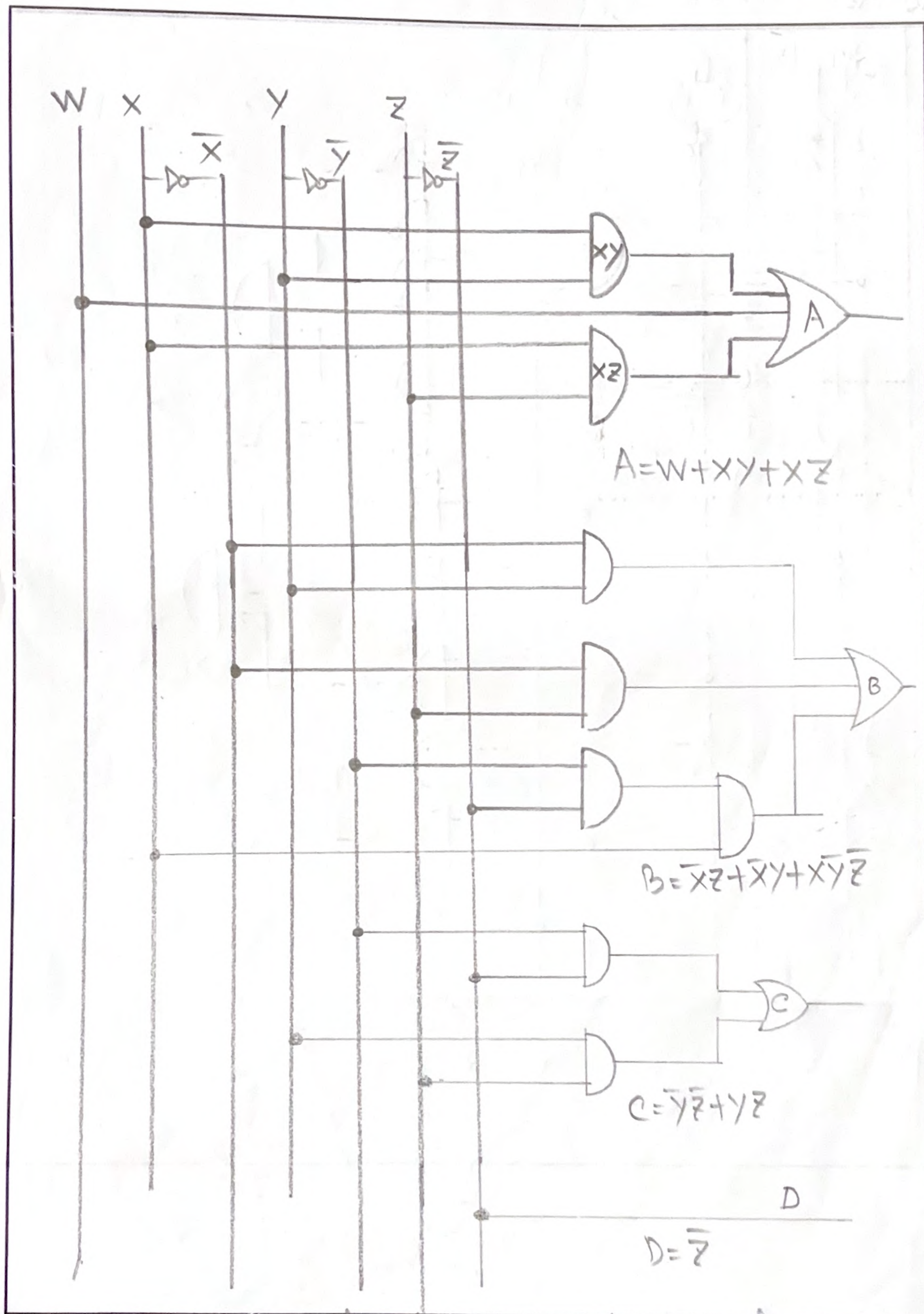
$B = \bar{X}Z + \bar{X}Y + X(\bar{Y}\bar{Z})$

WX \ YZ	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	X	X	X	X
10	1	0	X	X

$$C = \bar{Y}\bar{Z} + YZ$$

$$D = \bar{Z}$$

Figure F1: K-Maps



Minimal 1st Canonical Circuit

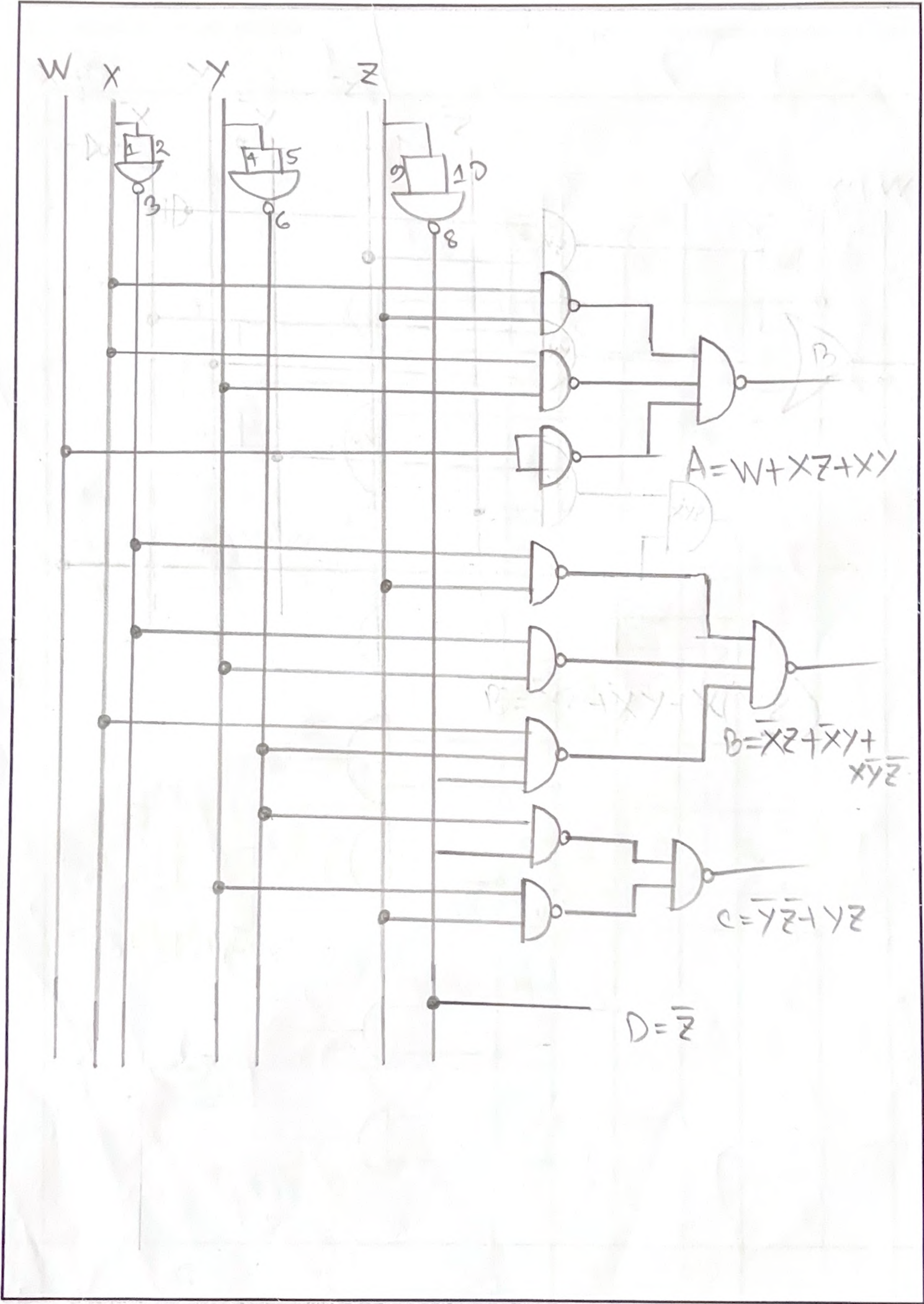


Figure 52: Minimal universal gate implementation of BCD to Excess-3 converter